MyungJin Ko (Ray)

Software Engineer · 7 Years

Seoul, South Korea | +82 10-2925-6426 | rayleighko2@gmail.com
github.com/rayleighko | linkedin.com/in/rayleighko | ray-k.medium.com

SUMMARY

Software engineer with 7 years of experience, specializing in multi-tenant platforms built for spiky, event-driven traffic. Architected a business-to-business multi-tenant Software-as-a-Service (SaaS) platform from **zero to production** — now used daily by **10+ business customers** — on infrastructure that absorbs **25× traffic spikes** (~1,000 baseline daily active users to ~25,000 concurrent) with **zero downtime and no over-provisioning**. Primary stack TypeScript, Next.js, and PostgreSQL, with production systems shipped in Rust, Java, and C# where the problem called for it.

EXPERIENCE

Plancy — Founding Engineer / Head of Engineering (Co-founder) May 2021 – Jun 2026

Owned architecture, development, infrastructure, and operations across two product lines for the MICE (meetings, incentives, conferences, exhibitions) industry, as the only experienced engineer on a 3-person team.

MICE All-in-One — Multi-Tenant SaaS Platform (mice.ai.kr) · Lead Engineer Feb 2024 – Jun 2026

- **Architected** a multi-tenant SaaS platform from proof of concept to production, now used daily by **10+ agency customers**, by modeling each *event* rather than each user as the tenant unit — aligning isolation, permissions, and cost on one axis so infrastructure cost scaled with projects, not end users.
- **Designed** infrastructure absorbing **25× traffic spikes (~25,000 concurrent users** vs. a ~1,000 baseline of daily active users) across 3–4 government-backed events per year, by pre-rendering read paths behind a content delivery network, tuning PostgreSQL indexes and connection pooling, and load-testing capacity pre-event — **zero downtime, no over-provisioning**.
- **Diagnosed** serverless timeouts in bulk document generation as the platform's worst bottleneck and migrated that path alone to a dedicated **Rust (Axum)** service through a strangler-fig migration, **eliminating batch job failures**.
- **Standardized** hand-built per-event registration forms into a **JSON-Schema-driven no-code form builder** customers configure themselves, and opened the platform to integrations through a public REST API (application programming interface) with scoped keys.
- **Integrated** three separate login flows into one identity model with per-event isolation via PostgreSQL Row-Level Security, and automated self-service domain provisioning through the Cloudflare API.
- **Built** the event-operations toolkit: real-time business matching with live chat and scheduling, QR-code on-site check-in with funnel analytics, and recurring invoicing and email with delivery tracking.
- **Established** release safety on a 2-engineer team through automated tests on critical flows, preview deployments, rollback procedures, and Sentry/PostHog instrumentation.
- **Mentored** the team's junior engineer from code review to independent feature ownership, and prototyped a retrieval-augmented generation proposal assistant validating the product's AI direction.

Stack · TypeScript, Next.js, React, Node.js, Rust (Axum), Supabase (PostgreSQL, RLS, Realtime), Cloudflare, Docker, Vercel, TipTap/YJS, Tauri, Sentry, PostHog, GitHub Actions

Event Registration Platforms — National Institutions · Technical Lead Oct 2021 – Oct 2025

- **Improved** delivery speed across **20–30 events per year** (each reaching ~2,000 daily active users at peak registration) over 4 years by abstracting recurring site and registration flows into reusable templates.
- **Collaborated** with stakeholders at national institutions (National Hangeul Museum, Sculpture City Seoul) to scope requirements, then delivered real-time registration platforms from design through production operations.
- **Operated** infrastructure, deployment pipelines, and the team's internal development environment and tooling.

Stack · Next.js, TypeScript, React, Node.js, Supabase/PostgreSQL, Vercel, Cloudflare, Docker

Systems & Integration Engineering · Software Engineer Oct 2021 – Oct 2025

- **Engineered** a solar-design review platform bridging desktop computer-aided design and the web: an Autodesk Revit add-in (C#/.NET) extracting building-information-modeling data into a custom file format (.pim), surfacing reports, charts, and national-standard compliance checks on Next.js and Supabase.
- **Implemented** a public-sector lookup system in Java and Spring on a government security-certified cloud (NHN Cloud), including encryption of personally identifiable information and access-control design.

Stack · C#/.NET, Autodesk Revit API, Java, Spring Boot, Next.js, TypeScript, NHN Cloud

TheScentLab — Founding Engineer (Co-founder) Sep 2020 – May 2021

- **Developed** and launched a personalized fragrance recommendation app (React Native, Expo) on iOS and Android, with a recommendation engine over a **~70,000-item dataset** combining Elasticsearch weighted scoring and survey-based personalization.
- **Deployed** the NestJS backend as containers on Amazon Web Services; **co-filed a patent** for the recommendation process and secured **~KRW 50M** in government startup funding.

Stack · React Native, Expo, TypeScript, NestJS, Elasticsearch, PostgreSQL, AWS (ECS, EC2, RDS, Lambda)

Supaja — Frontend Lead (Contract) Apr 2020 – Sep 2020

- **Created** the service's entire frontend as the sole engineer and handed it over with structure and documentation so it ran without further involvement; coached one developer new to React.

Weplanet — Web Developer Intern Mar 2019 – Aug 2019

- **Built** web prototypes across client projects — React frontends with Node (Express) and MySQL backends.

PROJECTS

Bearings — Solo Engineer (thebearings.app) Jun 2026 – Present

A regime-analysis tool that reads which economic regime a given portfolio is betting on — no login, no brokerage linking, no stored data. Source: github.com/rayleighko/thebearings.app

- **Built** and operate a Progressive Web App on Next.js 16, React 19, and TypeScript as a team of one, including the macroeconomic data pipeline (Bank of Korea ECOS, Federal Reserve Economic Data), observability, and deployment.
- **Designed** a 3-layer safety filter enforcing the boundary between decision support and investment advice at the system level for Claude-powered briefing and chat.

Stack · Next.js 16, React 19, TypeScript, Supabase, Claude API, Tailwind, PostHog, Sentry, Vercel

SKILLS

Primary	TypeScript, React, Next.js, Node.js, NestJS, PostgreSQL, Supabase
Worked with	Rust (Axum), Java/Spring, C#/.NET, React Native, Tauri, Express, Elasticsearch, MySQL
Infrastructure	Amazon Web Services (ECS, EC2, RDS, Lambda), Docker, Vercel, Cloudflare, NHN Cloud, GitHub Actions, Sentry, PostHog

EDUCATION & OTHER

Education	Sahmyook University — Bachelor of Science in Computer Science (Software major), Feb 2021
Patent	Co-filed, personalized recommendation process (TheScentLab)
Military Service	Republic of Korea Marine Corps, information technology operations, Feb 2014 – Nov 2015
Languages	Korean (native) · English (professional reading and writing)